Koushik Padmanabhan

Los Angeles, CA | +1 213-522-5710 | koushikkp2003@gmail.com | huggingface.co/koushik2k3
Portfolio: koushik2k3.github.io | linkedin.com/in/koushik-koli-padmanabhan | github.com/koushik2k3 | kousa2k3.itch.io

EDUCATION

University of Southern California Aug 2026 – May 2028
M.S. Computer Science (Game Development) *Los Angeles, CA*

Sri Ramachandra Institute of Higher Education and Research Aug 2021 – May 2025
B.Tech, Computer Science (AI & ML), CGPA 9.09/10, First Class with Distinction *Chennai, India*

TECHNICAL SKILLS

Languages: C++, C#, Python, JavaScript, SQL, Bash
Game Engines: Unreal Engine 5 (Blueprints), Unity (C#), Godot
Game AI & ML: Unity ML-Agents, Stable-Baselines3, PyTorch, TensorFlow, scikit-learn, XGBoost, OpenCV, Hugging Face Transformers
Tools: Git, Docker, FastAPI, Flask, React, Node.js, MongoDB, MySQL, MSSQL, n8n

GAME AI & GAMEPLAY PROJECTS

Unreal Traffic Simulator | Unreal Engine 5, Blueprints Demo video

- Built a multi-agent traffic simulation with Blueprint-driven vehicle AI: cars follow authored two-lane spline paths and brake when another vehicle enters their forward detection range, avoiding collisions without scripted waypoint timing.
- Implemented lane re-acquisition so vehicles spawned or displaced off-road find the nearest valid lane and merge back into traffic, keeping the simulation stable under arbitrary spawn placement.

Reinforcement Learning Agents | Unity ML-Agents, Stable-Baselines3, PyTorch huggingface.co/koushik2k3

- Trained Unity ML-Agents policies for three game tasks: physics-based locomotion and fetching (Huggy), positioning and firing at a target (SnowballTarget), and sparse-reward two-stage navigation (Pyramids), where a curiosity-driven intrinsic reward was needed to discover the button-then-pyramid sequence. All three run in the browser.
- Published 7 agents to the Hugging Face Hub with model cards and evaluation scores, covering tabular Q-learning (FrozenLake, Taxi), DQN learning from raw pixels (Atari Space Invaders), and PPO (LunarLander and the ML-Agents tasks).

Gameplay Prototypes | Unity (C#), Godot Play on itch.io

- Built 8 playable prototypes: an endless canyon flyer with randomized obstacle spawning, a wave-based physics arena where enemy count doubles each wave, a push-the-ball football game, a Fruit Ninja-style slicer, a side-scrolling runner, two shooter prototypes, and a 2D pixel-art platformer released on itch.io.
- Implemented the core systems in each: player input, spawn and difficulty pacing, physics-based pushing and collision handling, and win and loss states.

PUBLICATION

FocusNet: Deep Learning Framework for Fundus Pathology Springer, CVR 2025

- Designed FocusNet, a DenseNet169 backbone with a 1x1 pixel-wise attention module that classifies fundus images as normal, cataract, glaucoma, or diabetic retinopathy at 95% accuracy on the 4,000-image Kaggle Eye Diseases dataset.
- Benchmarked 12 configurations across ResNet, MobileNet, and DenseNet; attention raised DenseNet169 accuracy from 90% to 95% and diabetic-retinopathy F1 from 0.79 to 0.92. Published in Computer Vision and Robotics (CVR 2025), Springer, pp. 455–467.

EXPERIENCE

Mindera Feb 2025 – Jun 2026
Software Engineer *Bangalore, India*

- Built XGBoost-based anomaly detection pipelines for a fraud detection service, screening 1,000 transactions a day at 97% precision.
- Led an internal onboarding assistant built on Spec Kit that answers new developers' codebase questions in natural language with source citations and scaffolds their first change, cutting ramp-up from knowledge transfer to first contribution by 50%.
- Rebuilt the new-developer Docker environment after a migration left artifacts missing, replacing manual database dump downloads with automated SQL setup scripts and reducing setup from 2 hours to 30 minutes.

Katpro Technologies Dec 2023 – Dec 2024
Software Developer (Part-time) *Remote*

- Architected a leasing automation platform that chains AI voice, email, and SMS agents to take renters from inbound call to signed lease with no human coordination.
- Built a full-stack clinical coding application (React, FastAPI, Ollama) that reads physician notes through OCR and uses a locally hosted LLM with retrieval-augmented generation to return ICD-10 and CPT billing codes.
- Created an OCR and LLM document pipeline (Python, React, MSSQL) that extracts purchase orders, invoices, and GRNs at 95% field-level accuracy, replacing 20 hours of manual entry per week.
- Automated resume screening with LLM-based scoring and role-based routing in n8n, cutting time-to-shortlist by 60%.